

# Low-resource entity resolution with domain generalization and active learning

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## ABSTRACT

Entity Resolution (ER), a fundamental task in data cleaning and integration, is critical in various fields such as healthcare, e-commerce, and social networks. Traditional ER methods are constrained by the need for substantial labeled samples and the challenge of generalization to unseen domains. To address the low-resource challenge in ER, a novel two-phase framework is proposed. Initially, we introduce the Domain Generalization Entity Resolution (DGER) framework, combining domain adversarial learning and simulated target learning to improve the generalization performance on unseen target domains. Subsequently, to further adapt to the target dataset, we present a novel active learning approach called Domain-Aware Uncertainty Active Learning (DUAL), for fine-tuning the DGER model with minimal annotation cost. DUAL manually annotates target domain samples that are highly uncertain and exhibit high divergence from the source, while assigning pseudo-labels to high-confidence samples. Experimental results on multiple real-world datasets demonstrate that our framework outperforms traditional ER methods in generalizing to unseen domains. Specifically, our DGER method outperforms the best-performing ER baseline in each task, achieving an average F1 score improvement of 9.02% across eight different test tasks. Moreover, within a limited annotation budget during the active learning phase, our DUAL fine-tuning strategy for the ER model outperforms uncertainty-based active learning techniques.

## 1. Introduction

Entity Resolution (ER), also known as entity matching, record linkage, or reference reconciliation, is a fundamental problem in data cleaning and integration [1]. The key idea is to identify whether two entity records from two relational tables refer to the same real-world entity. Over the past decades, numerous ER methods have been proposed, ranging from rule-based methods such as disjunctive normal form [2], and general boolean formula [2], to machine learning-based methods such as SVM [3], and random forests [4], and deep learning-based methods such as DeepMatcher [5] and DeepER [6].

In recent years, ER has increasingly been addressed as a sequence classification task by fine-tuning pre-trained language models (PLMs), such as BERT [7], RoBERTa [8] and DistilBERT [9]. In particular, Ditto [10] serves as an effective demonstration of how PLMs can be adapted to ER tasks. However, this approach requires an extensive amount of high-quality labeled samples — often in the thousands — for typical ER applications. The complexity and diversity of real-world data make obtaining large-scale annotated ER data prohibitively expensive,

which makes ER a challenging problem in many practical scenarios, especially in low-resource settings. Two main strategies, transfer learning [11–15] and active learning [15–17], have been deployed to address the annotation budget problem in ER. Transfer learning employs previously labeled data from related domains to guide the learning process, reducing the need for extensive labeling of data in the target domain. On the other hand, active learning selectively annotates the most informative samples based on certain criteria, such as uncertainty, efficiently utilizing the annotation budget to improve the learning performance. However, current applications of these strategies present two fundamental challenges:

**Challenge I: Generalization to unseen domains:** While transfer learning has proven valuable in ER applications, its application scope remains limited, primarily focusing on domain adaptation [18]. This method exploits both labeled source domain data and unlabeled target domain data, essentially ‘seeing’ the target domain. However, in many real-world ER scenarios, obtaining sufficient data from the target domain itself can be prohibitively expensive or even infeasible. In such

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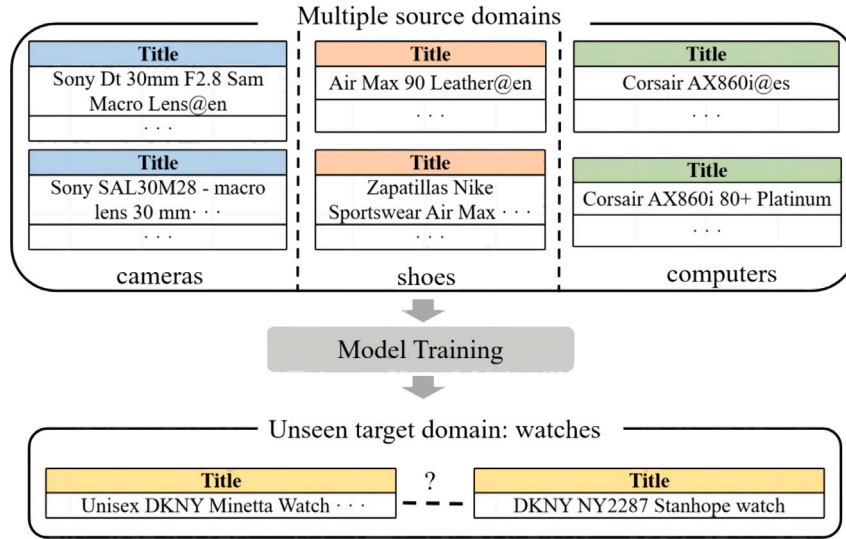


Fig. 1. An ER challenge: Training a model on multiple source domains (cameras, shoes, computers) with the expectation of competent performance on an unseen target domain (watches).

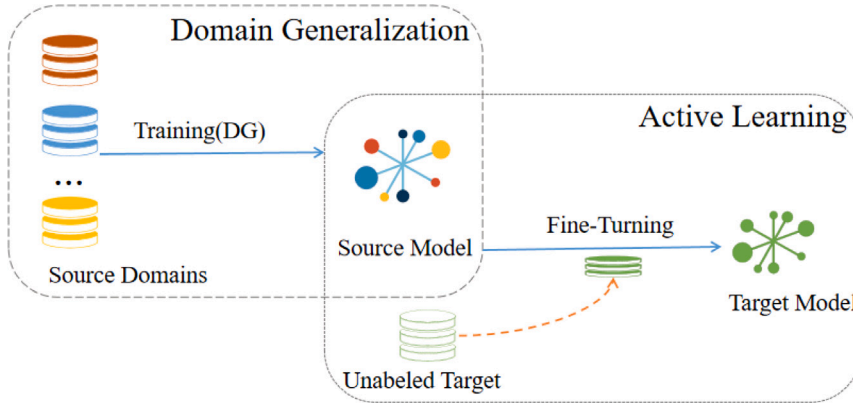


Fig. 2. The two-phase ER overview. The upstream (left) trains on source domains, generating a generalized model for unseen domains. The downstream (right) fine-tunes this model using active learning, applied to specific target domain, improving the model's performance within limited annotation budget.

cases, domain adaptation techniques that rely on the availability of target domain data become impractical. Fig. 1 illustrates this challenge, where an ER model is trained on multiple source domains such as cameras, shoes, and computers, and then expected to perform well on an unseen target domain like watches.

**Challenge II: Active learning amidst domain divergence:** Traditional active learning approaches for ER [15] focus on selecting the most informative samples for annotation based on uncertainty measures. These approaches, however, overlook the crucial aspect of divergence between target and source domains. Neglecting this divergence could lead to the exclusion of high-divergence samples, which are more likely to contain unique information for adapting to the target domain, thereby undermining potential performance improvements on the target domain, particularly under constrained annotation budgets.

To overcome these challenges, this paper introduces a two-phase framework for ER on unseen domains, combining upstream domain generalization and downstream active learning. Fig. 2 provides an overview of the work outlined in this paper.

**Addressing Challenge I:** In the upstream, we propose a novel approach to ER that leverages domain generalization (DG) [19–21] techniques to enhance the model's generalization performance on unseen target domains. The generalization of the model is crucial for practical deployment in real-world scenarios where labeled data may be limited or unrepresentative. Our Domain Generalization Entity Resolution (DGER) framework combines two learning strategies: domain

adversarial learning (DAL) [18] and simulated target learning (STL). DAL encourages the model to learn domain-invariant feature representations, enhancing its generalization to domain shifts. Meanwhile, STL, inspired by meta-learning [22,23] techniques, simulates domain shifts during the training phase by randomly selecting a source domain as a simulated target domain, enabling the model to learn more transferable knowledge from the source domains and better generalize to the target domain.

**Addressing Challenge II:** In the downstream, we present the Domain-Aware Uncertainty Active Learning (DUAL), a novel active learning-based approach for fine-tuning the DGER model. Unlike traditional active learning employed in ER, DUAL extends its focus beyond sample uncertainty by considering sample divergence from the source domains. This consideration enables the selection of target domain samples that diverge significantly from the source domains, which are more likely to yield higher performance gains. At the same time, in order to minimize the annotation budget and effectively utilize the large amount of unlabeled data, DUAL also selects high-confidence samples and assigns model-generated pseudo-labels to them, which further improves the performance of the model on the target domain.

**Contributions:** The contributions of this paper are threefold:

- We propose DGER, a novel ER approach that introduces domain generalization techniques for superior generalization performance on unseen target domains. DGER innovatively combines two key

strategies: domain adversarial learning, which learns domain-invariant representations, and simulated target learning, which simulates domain shift during training, to address the challenge of ER when the target domain is unseen. To the best of our knowledge, this is the first attempt to explicitly incorporate DG into the ER research.

- We propose DUAL, a domain-aware uncertainty active learning strategy for fine-tuning our DGER model. DUAL extends traditional active learning in ER by considering both sample uncertainty and divergence from the source domains, enabling efficient adaptation to the target domain with minimal annotation cost.
- Our experiments on multiple real-world datasets demonstrate that our proposed framework outperforms existing state-of-the-art ER methods in terms of generalization to previously unseen domains. Moreover, our active learning strategy achieves a significant reduction in the number of labeled samples required for training, making it a practical and cost-effective solution for real-world scenarios, particularly in low-resource settings.

## 2. Related work

### 2.1. Entity resolution

Entity Resolution (ER) is a fundamental task in data management that involves identifying records from one or more datasets that refer to the same real-world entity. Given two collections of records  $E_A$  and  $E_B$ , ER classifies a pair of entities  $(e_1, e_2), \forall e_1 \in E_A, e_2 \in E_B$  into match or non-match. This task has evolved through various methodologies, including rule-based [24,25], crowdsourcing-based [26–28], and machine learning-based [29–31] methods. Recently, deep learning approaches to ER [32–34] have shown promise, but the need for extensive labeled datasets remains a critical challenge. To reduce the dependence on high-quality training data, various strategies have been employed. Notably, leveraging transformer-based PLMs [10,35–37] has demonstrated strong performance in ER tasks.

In recent years, transfer learning has emerged as a promising avenue for enhancing ER performance. For instance, DADER [13] presents a domain adaptation-based ER framework, and TransER [38] uses an instance selector to pick source instances that have high confidence and bear a structural resemblance to the target domain. Despite these advancements, a key limitation of existing methods is their focus on domain adaptation, with insufficient consideration for generalization to unseen target domains. This limitation hinders their performance on unknown domains, restricting their applicability in diverse real-world scenarios.

### 2.2. Domain generalization

Domain generalization [19] is more challenging than domain adaptation, primarily because it does not rely on prior knowledge about the target domain. Although the methodological tools employed vary widely, existing domain generalization methods can generally be classified into three categories: data manipulation [39,40], representation learning [18,41], and learning strategies [20,21,42]. Data manipulation methods focus on enhancing the training data through augmentations and transformations, which include data augmentation and data generation techniques. However, these methods have limitations, such as the risk of overfitting to the artificially modified training data. Representation learning methods aim to learn domain-invariant features to enable models to adapt well to different domains. Learning strategy methods utilize advanced machine learning strategies, such as meta-learning [20] and ensemble learning [21,42], to enhance generalization across multiple training domains.

Our work makes a significant and unique contribution to the domain generalization field by conceptually introducing it into ER. By learning domain-invariant features, an ER model is built to generalize well to

unseen domains. Further, inspired by the meta-learning paradigm, our approach facilitates a model's quick adaptation to new domains, effectively solving ER problems even with limited or no prior knowledge of the target domain. By combining the advantages of these distinct, yet complementary, fields, we not only propose a novel ER approach, but also contribute to the domain generalization literature, setting the stage for future cross-disciplinary explorations.

### 2.3. Active learning

Active learning efficiently minimizes annotation effort in supervised learning, by iteratively selecting unlabeled data instances for manual labeling to maximize model performance. Approaches such as uncertainty sampling [43], query-by-committee [44], core-set [45], and expected model change maximization [46] have been utilized to guide this selection process.

Deep learning has shown exceptional performance in various applications in recent years, but it relies on large labeled datasets. To mitigate this problem, active learning has been successfully integrated with deep learning in recent studies, including text classification [47], image recognition [48], and object detection [49]. When applied to ER, active learning has shown promise in improving annotation efficiency. However, previous active learning approaches for ER [15,50,51] have not adequately considered sample divergence from the source data, which is an essential factor that may limit their effectiveness when applied to ER tasks with significant domain differences. Our work addresses this challenge by presenting an approach that considers this divergence during the sample selection process, potentially leading to improved active learning efficiency in such scenarios.

## 3. Problem statement

ER is the process of identifying and linking different data entries that pertain to the same real-world entity. To elucidate, consider two datasets,  $E_A$  and  $E_B$ , each comprising records that can be abstracted as a collection of key-value pairs  $\{(\text{attr}_i, \text{val}_i)\}_{1 \leq i \leq k}$ . Here,  $\text{attr}_i$  and  $\text{val}_i$  respectively represent the attribute name and its corresponding value, encapsulating the characteristics of the entities depicted in these records.

The primary objective of ER, as shown in Fig. 3, is to classify pairs of records  $(e_1, e_2), \forall e_1 \in E_A, e_2 \in E_B$ , assigning a label  $y$ , where  $y = 1$  indicates a match (both entries refer to the same real-world entity) and  $y = 0$  indicates a non-match. This classification is grounded in either a predefined labeling function or established ground truth, aiming to accurately reflect the underlying reality of entity identity.

Given the vast number of potential record pairs between  $E_A$  and  $E_B$ , the computational cost of ER process grows quadratically, necessitating an initial step known as blocking. This step significantly narrows down the comparison field by filtering out pairs unlikely to match, thus forming a subset  $C \subset E_A \times E_B$  with  $|C| \ll |E_A \times E_B|$ . Following blocking, the matching step evaluates each pair  $(e_1, e_2) \in C$ , focusing on their similarity to determine if they represent the same entity. In this paper, our primary focus is on the matching step, which can be effectively solved by formulating it as a sequence classification task.

## 4. Domain generalization for entity resolution

In this section, we introduce our proposed method, Domain Generalization for Entity Resolution (DGER), an ER approach designed to overcome the challenges of applying ER models to unseen domains. We first provide a detailed description of the DGER framework, and then we focus on the training stage where the model learns from multiple domains to improve its generalization capacity. Finally, we elucidate the inference stage where the trained model is applied to unseen data from the target domain to predict entity pair matches.

| Title                   | Author             | Venue        | Year |   | Title                     | Author             | Venue            | Year |
|-------------------------|--------------------|--------------|------|---|---------------------------|--------------------|------------------|------|
| On-demand data...       | p triantafillou... | vldb         | 1997 | ✓ | On-demand data...         | p triantafillou... | proc. of 23rd... | 1997 |
| Banks : browsing and... | b aditya...        | vldb         | 2002 | ✗ | Distance browsing...      | gr hjaltason...    | acm tran...      | 1999 |
| Path sharing...         | y diao...          | acm trans... | 2003 | ✓ | Yfilter : path sharing... | y diao...          | acm tran...      | null |

$E_A$ 
 $E_B$

Fig. 3. ER: determine the matching entries from two datasets.

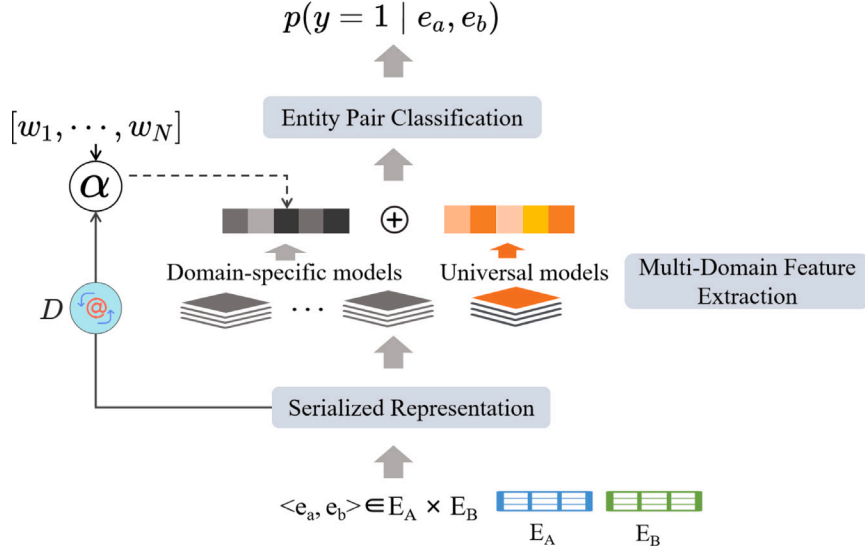


Fig. 4. The Domain Generalization for Entity Resolution (DGER) architecture.

#### 4.1. DGER framework

The DGER framework, as shown in Fig. 4, consists of three core modules: serialized representation, multi-domain feature extraction, and entity classification. The serialized representation is the pre-processing module that serializes the input entity pairs into a uniform form. Subsequently, the multi-domain feature extraction module integrates multiple PLMs to perform feature extraction and aggregation on the input entity pairs, thus obtaining a comprehensive feature representation. Finally, the entity classification module processes this feature representation to produce matching results. In the following, the DGER framework is described in detail.

##### 4.1.1. Serialized representation

The serialized representation module is responsible for pre-processing the entity pairs. The serialization of the entries is encoded according to Ditto [10] by serializing and concatenating all attributes/values of both entries into a sequence:

$$r_{e_i} = [\text{COL}] \text{attr}_1 [\text{VAL}] \text{val}_1 \dots [\text{COL}] \text{attr}_k [\text{VAL}] \text{val}_k \quad (1)$$

where [COL] and [VAL] are special tokens that denote the start of attributes and values, respectively. The input of ER is a pair of records  $x = (e_a, e_b)$ .

Thus, the process of taking a pair of entries as input and producing a serialized pair of entries, denoted by  $R$ , is as follows:

$$R(x) = R((e_a, e_b)) = [\text{CLS}] r_{e_a} [\text{SEP}] r_{e_b} [\text{SEP}] \quad (2)$$

where [SEP] and [CLS] are BERT special tokens that are added into the sequence similar to the sentence pair classification setting. Finally, the resulting sequence pair is given as input to the next module.

##### 4.1.2. Multi-domain feature extraction

The multi-domain feature extraction module is composed of  $N + 1$  DistilBERT models, including  $N$  domain-specific models denoted by  $\{f_{S_i}\}_{i=1}^N$ , and a universal model denoted by  $f_u$ . Each domain-specific model, trained separately on a specific source domain, learns domain-specific features. In contrast, the universal model, trained on data from all source domains, learns domain-general features. By integrating both types of models, our proposed framework can capture both domain-specific and domain-invariant features, resulting in a more generalizable feature representation.

In practice,  $R(x)$  is used as input to the  $N + 1$  models to extract  $N$  source domain-specific embeddings, denoted by  $f_{S_i}(R(x)), i = 1, \dots, N$ , and a domain-universal embedding, denoted by  $f_u(R(x))$ . The embeddings from the source domain models and the universal model are extracted by utilizing the hidden state of the [CLS] token from the last Transformer block in each DistilBERT model, effectively leveraging the contextualized features captured by the DistilBERT model.

For notation simplicity, we introduce  $F_u(x)$  to denote the universal-based feature obtained by  $f_u(R(x))$ , and  $F_{S_i}(x)$  to denote the  $i$ -th source-specialized feature extracted by  $f_{S_i}(R(x))$ . Hence, the output of the feature extraction module, denoted by  $F$ , is defined in Eq. (3):

$$F(x) = \left\{ F_{S_i}(x) \right\}_{i=1}^N \cup F_u(x) = F_s(x) \cup F_u(x) \quad (3)$$

where  $F$  extracts  $N$  specialized embeddings and a universal embedding from the sequence pair. The aggregation of these specialized embeddings, represented by  $F_s(x)$ , is guided by the weights provided by the domain prediction branch in the subsequent inference phase.

To optimally guide the aggregation of these specialized embeddings, represented by  $F_s(x)$ , we incorporate a domain prediction branch in the subsequent inference phase. This branch, denoted as  $D$  in Fig. 4, serves

as a classifier that predicts the domain of the input data. The weights it provides dynamically adjust the contribution of each domain-specific model in the final prediction.

#### 4.1.3. Entity classification

The entity classification module concatenates  $F_s(x)$  and  $F_u(x)$  obtained from the previous module and uses the neural network layer for the final decision. Specifically, the feature vectors are fed into the fully connected layer and subsequently the matching probability  $p(y = 1 | e_a, e_b)$  is calculated by the Softmax function, where  $y = 1$  indicates that  $e_a$  and  $e_b$  refer to the same entity.

The output of the entity pair classification, denoted by  $E$ , is given by Eq. (4):

$$E(x) = \begin{cases} 1, & \text{if } p(y = 1 | e_a, e_b) \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

In Eq. (4),  $E(x)$  represents the binary classification result for the input entity pair  $x$ , where 1 indicates that the entities  $e_a$  and  $e_b$  are matched, and 0 indicates that they are not matched.

#### 4.2. Training stage

The DGER training process consists of two main parts: the joint training and the domain prediction branch. The former aims to train a domain generalization model that performs well in unseen domains. The latter aims at the inference phase, which can better guide the aggregation of the output of the domain-specific models.

##### 4.2.1. Joint training process

In the DGER framework, the training process is driven by optimizing a combined loss function, which is comprised of four individual loss components: domain-specific loss ( $L_S(X)$ ), universal model loss ( $L_U(X)$ ), domain adversarial learning loss ( $L_{DAL}(X)$ ), and simulated target learning loss ( $L_{STL}(X)$ ). The overall loss function is defined as the sum of these four components, and the training process aims to minimize this combined loss.

The combined loss function for the DGER framework is defined in Eq. (5):

$$L_{\text{total}}(X) = L_S(X) + L_U(X) + L_{DAL}(X) + L_{STL}(X) \quad (5)$$

In Eq. (5),  $L_{\text{total}}(X)$  represents the total loss to be minimized during training. Each component ( $L_S(X)$ ,  $L_U(X)$ ,  $L_{DAL}(X)$ , and  $L_{STL}(X)$ ) targets a specific aspect of our model's training, ensuring a comprehensive and effective learning process.

Given  $N$  labeled source domains  $\{S_i\}_{i=1}^N$ , a batch  $X$  is defined as follows:

$$X = \{(x_i^{S_j}, y_i^{S_j}) | 1 \leq i \leq |S_j|, 1 \leq j \leq N\} \quad (6)$$

where  $(x_i^{S_j}, y_i^{S_j})$  represents the  $i$ th instance and its corresponding label from  $S_j$ , and  $|S_j|$  denotes the number of instances sampled from the  $j$ th source domain in the current batch. The set  $X$  contains pairs of instances and their labels from all source domains in the current batch.

**Domain-specific and universal training.** For each source domain  $S_i$ , we train a domain-specific model  $f_{S_i}$  using the available labeled data from that domain. These models are tailored to capture domain-specific features and patterns, thereby improving performance within the respective domains.

An entity classification module  $E_{S_i}$  is added to each  $f_{S_i}$  to predict the probabilities of matching and non-matching for each domain  $S_i$ . The training loss for the corresponding domain-specific  $L_S$ , defined in Eq. (7), is:

$$L_S(X) = \sum_{i=1}^N \sum_{x \in S_i} L_{CE}(E_{S_i}(F_{S_i}(x^{S_i})), y^{S_i}) \quad (7)$$

Here,  $L_{CE}$  denotes the cross entropy loss function, which measures the difference between the predicted and true domain labels. All subsequent loss functions in this article will be based on the cross-entropy loss function.

In conjunction with domain-specific models training, a universal model  $f_u$  is trained to learn domain-invariant features across all source domains. The universal model is trained using the same loss function and optimization algorithm as the domain-specific models but utilizes data from all source domains during training.

The universal model  $f_u$  is optimized by adding an entity classification module  $E_u$  to predict the probabilities of matching and non-matching for all source domains. The universal model utilizes a loss function  $L_U$ , defined in Eq. (8).

$$L_U(X) = \sum_{i=1}^N \sum_{x \in S_i} L_{CE}(E_u(F_u(x)), y) \quad (8)$$

Here,  $x$  represents an arbitrary input from source domains, and  $y$  denotes the corresponding output label.

**Combined learning strategies.** In our approach, we combine domain adversarial learning (DAL) and simulated target learning (STL) to enhance the model's performance on tasks of the target domain. DAL is employed to learn a domain-invariant feature representation that is insensitive to the different data distributions. Meanwhile, STL, inspired by meta-learning principles, helps the model learn transferable knowledge and better generalize to the target domain by simulating domain shifts during training.

##### (1) Domain Adversarial Learning (DAL)

We employ the Gradient Reversal Layer (GRL) [52] method for domain adversarial training to obtain domain-invariant features from  $F_u$ , facilitating smoother transfer to the target domain. The DAL strategy involves optimizing a generator  $F_u$  with learnable parameter  $\theta_u$  and a discriminator  $d$  with learnable parameter  $\theta_d$  using a min-max objective function, similar to generative adversarial networks (GANs) [18].

The GRL method functions by reversing the gradients during the backward pass of the neural network, encouraging  $F_u$  to generate features that challenge  $d$ 's ability to distinguish between domains. Meanwhile,  $d$  strives to classify samples into their respective domains. Let  $\lambda$  be a scalar hyperparameter that controls the weight of the domain adversarial loss. The GRL can be defined as a function  $G$ , which is applied to the input  $x$  as follows:

$$G(x) = x, \quad G'(x) = -\lambda \quad (9)$$

In Eq. (9),  $G(x)$  and  $G'(x)$  represent the forward and backward pass of the GRL, respectively. During the backward pass, the gradient is multiplied by  $-\lambda$ , effectively reversing the gradient flow and encouraging the generator to produce domain-invariant features. The optimization for the DAL is formulated as a min-max problem, aiming to find the optimal parameters for both the generator and the discriminator:

$$\min_{\theta_d} \max_{\theta_u} F(d, F_u, \theta) \quad (10)$$

Here,  $F(d, F_u, \theta)$  is the objective function, where  $\theta$  consists of  $\theta_d$  and  $\theta_u$ , the respective parameters of  $d$  and  $F_u$ . The goal is to minimize  $d$ 's loss in distinguishing domains and maximize  $F_u$ 's loss in generating domain-invariant features. The objective functions for updating  $d$  and  $F_u$  are as follows:

$$F(d, F_u, \theta) = \sum_{i=1}^N \sum_{x \in S_i} L_{CE}(d(H(F_u(x^{S_i})), \theta)), i) \quad (11)$$

where  $H(x, \theta) = \begin{cases} x, & \text{if } \theta = \theta_d \\ G(x), & \text{if } \theta = \theta_u \end{cases}$

In Eq. (11),  $H(x, \theta)$  applies  $G(x)$  to the input feature  $x$  when updating the generator parameters  $\theta_u$ , and keeps  $x$  unchanged when updating the discriminator parameters  $\theta_d$ .

Finally, the total loss function  $L_{\text{DAL}}(X)$  is combined in Eq. (12):

$$L_{\text{DAL}}(X) = F(d, F_u, \theta_d) - \lambda F(d, F_u, \theta_u) \quad (12)$$

By minimizing  $L_{\text{DAL}}(X)$ , the parameters of both the generator and the discriminator are simultaneously optimized.

#### (2) Simulated Target Learning (STL)

STL is inspired by meta-learning [22,23] techniques, which have been proven effective for achieving domain generalization. In our proposed method, domain shifts are simulated during the training phase to create a model that can generalize well to unseen target domains. Specifically, the original data is organized into source domains, and for each learning iteration, one of the  $S$  source domains is randomly selected as the simulated target domain  $S_{\text{sim}}$ . The remaining  $S - 1$  domains serve as the source domains for that iteration, represented by  $S_{\text{src}}$ . This simulates the shifts between the true training and testing domains, enabling the model to perform well on the final testing evaluation on the target domain  $T$ .

Let  $F_{\text{sim}}$  denote the process of combining the features from all source domains (excluding the simulated target) and the universal model. These combined features are then fed into the entity classification module  $E$  for further processing, as detailed in Eq. (13):

$$F_{\text{sim}}(x, \text{sim}) = \left\{ F_{S_i}(x) \right\}_{i=1, i \neq \text{sim}}^K \cup F_u(x) \quad (13)$$

Eq. (13) is applied for each batch during the training process, with the simulated target domain  $\text{sim}$  being randomly selected for each iteration. Finally, the simulated-target loss  $L_{\text{STL}}$  for the batch  $X$  is given by:

$$L_{\text{STL}}(X) = \sum_{i=1}^N \sum_{x \in S_i} L_{CE}(E(F_{\text{sim}}(x, \text{sim})), y) \quad (14)$$

Eq. (14) quantifies the effectiveness of our STL process, emphasizing its importance in training the model to generalize well to new, unseen domains.

In summary, our STL method enhances generalization through simulating domain transfer, ultimately improving performance on unseen target domains.

#### 4.2.2. Domain prediction branch

The domain prediction branch is trained separately from the core modules of DGER. The classifier is trained to predict the domain label of a given input based on these features. The loss function for the domain prediction branch training,  $L_D(X)$ , is defined in Eq. (15):

$$L_D(X) = \sum_{i=1}^N \sum_{x \in S_i} L_{CE}(D(x), i) \quad (15)$$

The domain prediction branch functions as a plugin for the inference stage, assessing the relevance of the input sample to the source domains used in the training stage and dynamically adjusting the weights of the domain-specific models.

#### 4.3. Inference stage

During inference, DGER is applied to unseen target data to predict entity pair matches. The domain prediction branch  $D$  provides weights, which map a given target domain entity pair  $x$  to each source domain. Specifically, given an input  $x$ , the mapping is computed as follows:

$$w = D(x) = [w_1, \dots, w_N]^T \quad (16)$$

These weights from Eq. (16) are subsequently used to calculate the aggregated feature representation  $F_s(x)$  from the domain-specific models, as follows:

$$F_s(x) = \sum_{j=1}^N w_j F_{S_j}(x) \quad (17)$$

In summary, the DGER framework's inference strategy involves a series of computations and transformations. It leverages both specialized and universal features, as well as a domain prediction branch, to effectively handle unseen target domains.

### 5. Fine-tuning DGER via active learning

While the DGER model exhibits generalization across unseen domains, achieving optimal performance on specific target domains remains a challenge. There may still be instances where the model encounters performance stagnation or difficulties in adapting to the unique characteristics of specific target domains. Therefore, to improve the performance of the model in a specific target domain with a limited annotation budget, we introduce an innovative approach to fine-tune the DGER model. We term this approach Domain-Aware Uncertainty Active Learning (DUAL), a method that ingeniously combines the generalization of DGER with the iterative learning process of active learning, with special consideration of the context of domain divergence, to address the challenge of adapting DGER to specific domains.

#### 5.1. DUAL: Domain-aware uncertainty active learning framework

The DUAL framework (illustrated in Fig. 5) combines the power of active learning with the DGER. This iterative process selects two types of samples from the unlabeled target domain data: high-informative samples, which are annotated by human experts, and high-confidence samples that receive pseudo-labels based on model's predictions. By leveraging expert knowledge and the DGER model's confidence, the DUAL framework propels DGER's adaptation to the target domain, while significantly reducing the annotation budget.

#### 5.2. Selection of high-informative samples

The selection of high-informative samples is motivated by the potential for improving the model performance, as they represent instances where the model is the most uncertain or less confident. These samples are selected for human annotation to leverage the knowledge of the annotators to provide more accurate and informative labels. To this end, we propose a novel criterion called domain-aware uncertainty, which takes sample uncertainty and the degree of sample source divergence into account.

(i) Uncertainty: Uncertain samples help to pinpoint areas where the model's prediction is less confident or lacks sufficient knowledge. These instances, usually close to the decision boundary, can provide valuable insights to refine the model's understanding, thus enhancing the predictive performance.

To identify sample uncertainty, we employ the entropy of the conditional probability distribution computed by the model at its current state of fine-tuning. Specifically, let  $p(x)$  be the probability that a record pair  $x$  is matched according to the current model. Then, the uncertainty of an entity pair  $x$  is quantified by the entropy  $H(x)$ , which is computed as follows:

$$H(x) = -p(x) \log p(x) - (1 - p(x)) \log (1 - p(x)) \quad (18)$$

(ii) Source Divergence: Identifying and focusing on samples with high divergence from the source domain is an innovative point of our approach. These divergent samples are crucial as they often embody unique characteristics of the target domain unseen in the source. In essence, these samples represent the 'new' knowledge that the model needs to acquire to adapt effectively to the target domain. Identifying and learning from those samples can yield significant performance gains, as they allow the model to more effectively adapt to the target domain.

To assess source divergence, we employ cosine similarity to measure the dissimilarity between the source and target domains. The average

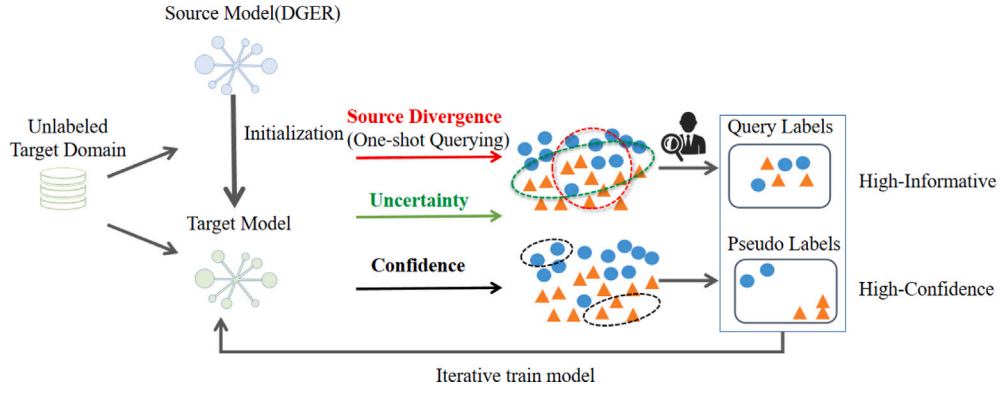


Fig. 5. The Domain-Aware Uncertainty Active Learning (DUAL) framework.

cosine similarity for each target domain sample  $x$  is computed with respect to all source domain samples  $X^S$  using the universal feature  $F_u$ . Lower cosine similarity values imply higher divergence from the source domain, computed as:

$$\text{AvgCosSim}(X^S, x) = \frac{1}{|X^S|} \sum_{x^S \in X^S} \frac{F_u(x^S) \cdot F_u(x)}{\|F_u(x^S)\| \|F_u(x)\|} \quad (19)$$

Eq. (19) details the method used to quantify the divergence of a target domain sample from the source domain. A one-shot querying strategy is applied in our method, whereby cosine similarity values are calculated once before the active learning process and these values are retained for all subsequent iterations. This strategy results in lower computational costs and quicker active learning iterations, providing a balance between computational resources and model performance.

In DUAL, the most informative samples are those with the highest domain-aware uncertainty. The domain-aware uncertainty score for a sample  $x$  is defined as the product of its entropy  $H(x)$  and its divergence from the source domain  $1 - \text{AvgCosSim}(X^S, x)$ . In each iteration, the top  $K$  pairs are selected from the pool of all unlabeled entity pairs  $Z^U$  as the most informative samples to be added to the training set for fine-tuning. This selection process can be formalized as:

$$\underset{Z \subseteq Z^U | |Z|=K}{\text{argmax}} \sum_{x \in Z} H(x) \cdot (1 - \text{AvgCosSim}(X^S, x)) \quad (20)$$

### 5.3. Selection of high-confidence samples

For high-confidence sample selection, we leverage the DGER to identify unlabeled target samples with high confidence scores. We assume that these samples are reliable and assign target-oriented pseudo-labels accordingly. This semi-supervised learning strategy effectively utilizes the information inherent in unlabeled samples.

The high-confidence sets are defined in Eq. (21) as:

$$\underset{Z \subseteq Z^U | |Z|=K}{\text{argmin}} \sum_{x \in Z} H(x) \quad (21)$$

Eq. (21) selects a subset  $Z$  from the unlabeled dataset  $Z^U$  by minimizing the total entropy of the selected samples. This corresponds to choosing samples with the highest prediction confidence, allowing the model to prioritize pseudo-labeling the most reliable samples for further training.

In summary, the DUAL approach enables the model to better adapt to the target domain by selectively including high-informative and high-confidence samples into the training set, thus effectively fine-tuning the DGER model and maximizing its performance in the presence of a limited annotation budget.

Table 1

Datasets for our experiments.

| Dataset        | Domain      | (#Train, #Valid) | #Test | #Attrs |
|----------------|-------------|------------------|-------|--------|
| Computers      | Product     | 2231, 536        | 1098  | 1      |
| Cameras        | Product     | 1409, 392        | 1057  | 1      |
| Watches        | Product     | 1756, 413        | 1012  | 1      |
| Shoes          | Product     | 1611, 404        | 1084  | 1      |
| Walmart-Amazon | Electronics | 6144, 2049       | 2049  | 5      |
| DBLP-ACM       | Citation    | 7417, 2473       | 2473  | 4      |
| Abt-Buy        | Product     | 5743, 1916       | 1916  | 3      |
| Amazon-Google  | Software    | 6874, 2293       | 2293  | 3      |

## 6. Experimental evaluation

### 6.1. Experimental setup

#### 6.1.1. Datasets

The experiments in this paper are conducted on eight different datasets, which are summarized in Table 1. These datasets are sourced from two benchmarks: the WDC [53] product matching benchmark and the Magellan [30] benchmark. They encompass a wide range of domains, including product, electronics, citation, and software. The composition of each dataset is a collection of candidate pairs of records, each from two structured tables with the same set of attributes.

This selection of datasets reflects the diversity and complexity of real-world data. The total number of attributes across all datasets varies from 1 to 5, and the proportion of matching entities ranges from 9.39% to 26.31%. Each dataset is partitioned into training, validation, and testing subsets. We adhere to the same splits as used in Ditto [10] for consistency and comparability.

#### 6.1.2. Evaluation metrics

The experimental evaluation involves the following standard metrics:

- **Precision (P)** is computed as the ratio of true positive predictions to the total number of positive predictions ( $P = \frac{TP}{TP+FP}$ ), where  $TP$  and  $FP$  represent the number of true positives and false positives, respectively.
- **Recall (R)** captures the proportion of actual positives correctly predicted as such, expressed as  $R = \frac{TP}{TP+FN}$ , with  $FN$  denoting false negatives.
- **Accuracy (Acc)** reflects the overall correctness of predictions across all cases, and is calculated by  $Acc = \frac{TP+TN}{TP+TN+FP+FN}$ , where  $TN$  is the count of true negatives.
- **F1 Score (F1)** is the harmonic mean of precision and recall, especially pertinent for tasks where a balance between the two is sought. It is calculated using  $F1 = 2 \cdot \frac{P \cdot R}{P+R}$ .

**Table 2**  
Hyperparameter settings.

| Parameter               | Value              |
|-------------------------|--------------------|
| Training Epochs         | 10                 |
| Initial Learning Rate   | 1e-5               |
| Learning Rate Scheduler | Linear with warmup |
| Batch Size              | 16                 |
| Max Sequence Length     | 256                |

Given the inherent imbalance in ER datasets, the F1 Score becomes particularly crucial. It effectively balances precision and recall, serving as the most comprehensive indicator of model performance in scenarios where non-matching entities are predominant. Thus, F1 is the focal metric in our experiments, in line with standard evaluation practices in ER research.

### 6.1.3. Environment and hyperparameters

Experiments were conducted on a server with an Intel Xeon Gold 6226R CPU (64 cores, 2.90 GHz), an NVIDIA GeForce RTX 3090 GPU (24 GB memory), and 1024 GB system memory, under Ubuntu 20.04 LTS. This setup enabled efficient deep learning model execution.

Hyperparameter settings are detailed in Table 2, applied uniformly across all experiments to ensure comparability and reproducibility.

All experiments were conducted five times to ensure result stability and reproducibility. The average of these repetitions is reported. This consistent approach across different methods aids in a fair comparison of generalization performance.

### 6.1.4. Baselines

In our comparison, DGER is evaluated against five notable ER methods. We adhere to the specifications outlined in the original papers and implement our baselines using the corresponding publicly available codes.

- DeepMatcher [5]: a pioneering work in deep learning-based ER, leverages a Recurrent Neural Network with an attention mechanism to create attribute vectors, which are then used by a binary classifier for ER.
- HierMatcher [54]: offers a hierarchical matching network for ER. It effectively handles heterogeneous and dirty data by implementing a cross-attribute token alignment and comparison layer, an attribute-aware attention mechanism, and an entity level matching layer.
- Ditto [10]: a pioneering ER methodology that first leverages PLMs, including BERT. The effectiveness of Ditto is further enhanced by incorporating domain knowledge, summarizing long strings, and including challenging samples in the training data.
- Rotom [35]: presents a meta-learning framework that selects and weighs augmented data to fine-tune PLMs more effectively.
- HierGAT [37]: combines the strengths of the PLMs and graph attention mechanisms to improve the performance of ER tasks by modeling the interdependencies between entities and attributes.

## 6.2. Experimental results

### 6.2.1. Overall performance of DGER

To assess DGER's performance, evaluations were conducted under two distinct settings: homogeneous domains and heterogeneous domains. These settings account for the structure and degree of domain shifts between source and target datasets. For each setting, we design four test tasks to evaluate the performance of DGER. Let  $D = \{D_1, D_2, D_3, D_4\}$  denote the set of four datasets in each setting. In the  $i$ th task ( $i = 1, 2, 3, 4$ ), the model is trained on the source domains  $D \setminus D_i$  and evaluated on the target domain  $D_i$ , which remains unseen during the training process.

- Homogeneous domains: In this scenario, we employ the datasets from the WDC product matching benchmark, where  $D = \{\text{Computers, Cameras, Watches, Shoes}\}$ . These datasets share the same attribute names but originate from different domains. Note that some baselines, DeepMatcher [5] and HierMatcher [54], are not applicable to heterogeneous datasets, so their results are only reported in the homogeneous dataset experiments.
- Heterogeneous domains: In this scenario, we utilize four datasets from the Magellan benchmark, where  $D = \{\text{Walmart-Amazon, DBLP-ACM, Abt-Buy, Amazon-Google}\}$ . These datasets originate from different domains, have distinct attribute names and counts, and exhibit similar scales. They represent a complex and realistic scenario of heterogeneous domain representations.

#### (1) Homogeneous domains

The DGER model demonstrates domain generalization capabilities in the context of homogeneous domains, outperforming other baselines. Table 3 presents the experimental results that affirm DGER's effectiveness in leveraging information embedded in source domains to generalize to unseen target domains.

#### (2) Heterogeneous domains

DGER shows commendable generalization to diverse dataset structures and representations when handling heterogeneous domains, outperforming Ditto [10], Rotom [35] and HierGAT [37] in all assessed tasks. The results in Table 4 underline DGER's potential in navigating complex, heterogeneous environments.

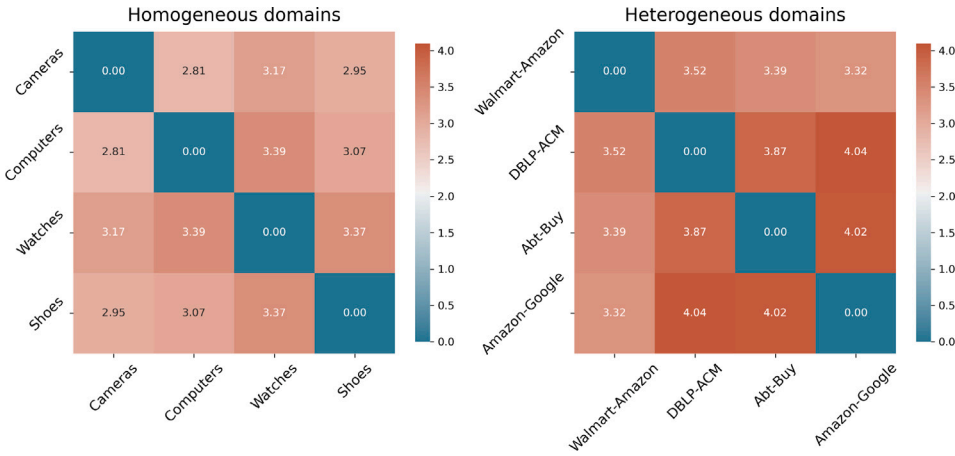
In particular, it can be observed that the HierGAT approach fails to be applied to all domains because of the limitations in dealing with the differences in the number of attributes across domains. Although the problem of inconsistency in the number of attributes between different source domains during the training phase is mitigated to a certain extent by the attribute padding strategy, this approach also inevitably introduces a certain amount of noise, which affects the generalization performance of the model. More importantly, when the number of attributes in the target domain exceeds the maximum number of attributes in source domains during the training phase, such as Walmart-Amazon in Table 4, this strategy cannot be implemented because we cannot arbitrarily eliminate the extra attributes in the target domain. This challenge further highlights the advantages of our approach in dealing with structurally heterogeneous datasets.

It is worth noting that despite DGER's superiority over baselines, its performance on certain heterogeneous domains, such as Walmart-Amazon, Abt-Buy, and Amazon-Google, is not sufficiently satisfactory for real-world applications. One potential factor limiting further generalization improvement may be the significant divergence in feature distributions between heterogeneous domains. To verify this hypothesis, we visualize the pairwise Maximum Mean Discrepancy (MMD) [55] distances between domains, as shown in Fig. 6. The MMD distance measures the difference between two probability distributions based on their kernel embeddings, with higher values indicating greater distributional divergence. The figure shows that the MMD distances between heterogeneous domains are generally larger than those between homogeneous domains, indicating more pronounced discrepancies in their feature distributions. Such distributional differences may imply an inherent performance bottleneck in the generalization to unseen domains under heterogeneous settings.

In conclusion, the overall results demonstrate DGER's superior generalization capabilities on unseen domains in both homogeneous and heterogeneous settings compared to baselines. Simultaneously, it has been noted that DGER's performance exhibits certain constraints in specific challenging domains, which may influence its direct applicability to real-world tasks. These findings motivate the exploration of active learning strategies to further improve its performance, which will be discussed in later sections.

**Table 3**  
Generalization performance on Homogeneous domains.

| Domain    | Metrics   | DGER          | HierGAT [37]  | Rotom [35]    | Ditto [10] | Hier-Matcher [54] | Deep-Matcher [5] |
|-----------|-----------|---------------|---------------|---------------|------------|-------------------|------------------|
| Cameras   | Precision | 0.8645        | 0.7717        | 0.7778        | 0.7082     | 0.5392            | 0.4781           |
|           | Recall    | 0.8896        | 0.8333        | 0.8507        | 0.8680     | 0.7396            | 0.8715           |
|           | Accuracy  | 0.9262        | 0.8874        | 0.8931        | 0.8666     | 0.7569            | 0.7058           |
|           | F1        | <b>0.8646</b> | 0.8013        | <u>0.8126</u> | 0.7800     | 0.6237            | 0.6175           |
| Computers | Precision | 0.7093        | 0.6189        | 0.6024        | 0.5562     | 0.4706            | 0.3776           |
|           | Recall    | 0.8896        | 0.8528        | 0.8361        | 0.8762     | 0.7759            | 0.8260           |
|           | Accuracy  | 0.8707        | 0.8169        | 0.8051        | 0.7759     | 0.7013            | 0.5820           |
|           | F1        | <b>0.7893</b> | <u>0.7173</u> | 0.7003        | 0.6805     | 0.5859            | 0.5184           |
| Watches   | Precision | 0.6798        | 0.5307        | 0.5168        | 0.5290     | 0.4791            | 0.3787           |
|           | Recall    | 0.9453        | 0.9453        | 0.9562        | 0.9306     | 0.7117            | 0.8321           |
|           | Accuracy  | 0.8646        | 0.7589        | 0.7460        | 0.7569     | 0.7125            | 0.5850           |
|           | F1        | <b>0.7908</b> | 0.6798        | <u>0.6851</u> | 0.6746     | 0.5727            | 0.5205           |
| Shoes     | Precision | 0.7008        | 0.5456        | 0.4880        | 0.5533     | 0.4773            | 0.4412           |
|           | Recall    | 0.8339        | 0.8915        | 0.9661        | 0.8610     | 0.8542            | 0.8915           |
|           | Accuracy  | 0.8579        | 0.7685        | 0.7149        | 0.7730     | 0.7057            | 0.6638           |
|           | F1        | <b>0.7616</b> | <u>0.6770</u> | 0.6485        | 0.6737     | 0.6124            | 0.5903           |



**Fig. 6.** Heatmaps of pairwise MMD distances between homogeneous and heterogeneous domains.

**Table 4**  
Generalization performance on Heterogeneous domains.

| Domain         | Metrics   | DGER          | HierGAT [37] | Rotom [35]    | Ditto [10]    |
|----------------|-----------|---------------|--------------|---------------|---------------|
| Walmart-Amazon | Precision | 0.2997        | N/A          | 0.2657        | 0.1524        |
|                | Recall    | 0.7720        |              | 0.8342        | 0.9326        |
|                | Accuracy  | 0.8087        |              | 0.7672        | 0.5051        |
|                | F1        | <b>0.4319</b> |              | <u>0.4030</u> | 0.2620        |
| DBLP-ACM       | Precision | 0.8623        | 0.6544       | 0.8556        | 0.7807        |
|                | Recall    | 0.9594        | 0.9977       | 0.5338        | 0.9864        |
|                | Accuracy  | 0.9652        | 0.9050       | 0.9001        | 0.9478        |
|                | F1        | <b>0.9083</b> | 0.7904       | 0.6574        | <u>0.8716</u> |
| Abt-Buy        | Precision | 0.5789        | 0.8967       | 0.4242        | 0.1583        |
|                | Recall    | 0.3738        | 0.5526       | 0.4078        | 0.5534        |
|                | Accuracy  | 0.9034        | 0.2039       | 0.8768        | 0.6356        |
|                | F1        | <b>0.4543</b> | 0.2979       | <u>0.4158</u> | 0.2462        |
| Amazon-Google  | Precision | 0.4763        | 0.4562       | 0.6519        | 0.3713        |
|                | Recall    | 0.6880        | 0.5342       | 0.4402        | 0.7094        |
|                | Accuracy  | 0.8909        | 0.8875       | 0.9189        | 0.8478        |
|                | F1        | <b>0.5629</b> | 0.4921       | <u>0.5255</u> | 0.4875        |

### 6.2.2. Ablation study on DGER

To further understand the efficacy of the components of the DGER model, we conducted an ablation study dissecting the two phases in DGER: the training phase and the inference phase.

#### (1) Training Phase

In the training phase, our primary goal was to evaluate the roles played by the DAL strategy and the STL strategy in the performance of DGER. To achieve this, we trained the model with these strategies

in isolation to gauge their individual and aggregate impacts. The F1 results are presented in Fig. 7.

As shown in Fig. 7, the full DGER model consistently outperforms the variants without DAL or STL across all domains, demonstrating the effectiveness of combining these two strategies. DGER (w/o DAL) achieves higher F1 scores than DGER (w/o STL) in 7 out of 8 domains. This might be because DAL contributes more to the model's performance by learning domain-invariant features, which is essential for generalizing to unseen domains. While STL improves the model's adaptability by simulating domain shifts during training, its effect may be limited if the learned features are not sufficiently domain-invariant. In some unique cases, such as the Abt-Buy domain, the effectiveness of STL might be compromised.

Overall, the ablation study suggests that the combination of DAL and STL is critical for DGER with each strategy playing a unique and complementary role in improving the model's generalization performance on unseen domains.

#### (2) Inference Phase

In the inference phase, we assessed the impact of the domain prediction branch on the model's ability to generalize to unseen domains. Three configurations were compared:

- DGER (w/ Weighting): This configuration utilizes a domain prediction branch to determine the weight of each domain-specific output before averaging.
- DGER (w/ Max-Voting): This configuration also employs the domain prediction branch, but selects the domain with the highest score and uses its output only.

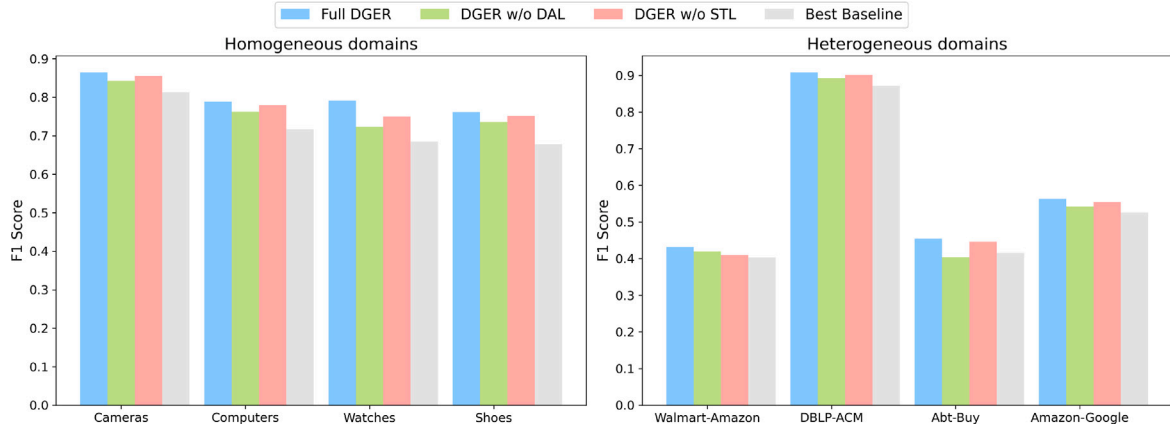


Fig. 7. Comparison of Different Training Strategies.

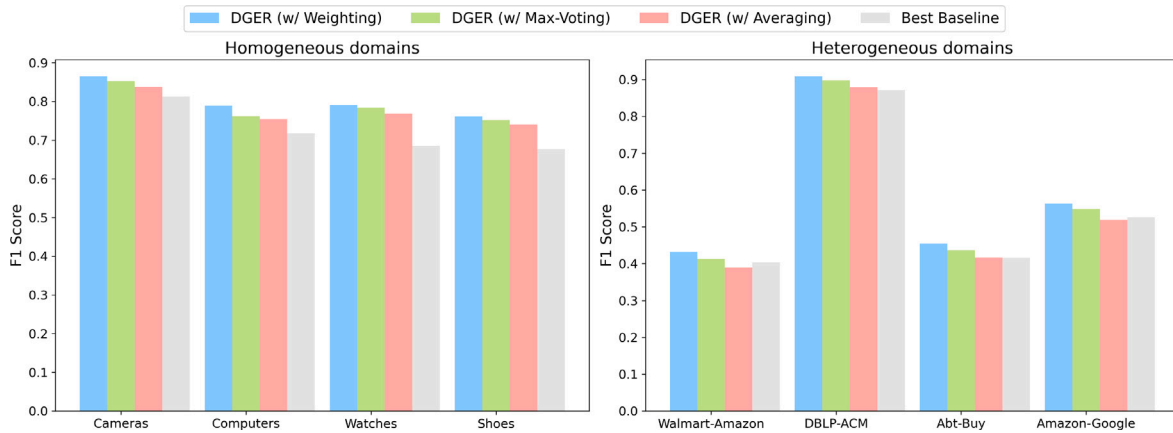


Fig. 8. Comparison of Different Inference Strategies.

- DGER (w/ Averaging): This configuration does not include a domain prediction branch and calculates the average of all domain-specific outputs directly.

The F1 results of this comparison are presented in Fig. 8. These results provide evidence that the chosen configuration, DGER (w/ Weighting), provides the best performance across the various domains.

The results in Fig. 8 indicate the superiority of DGER (w/ Weighting) across all domains, suggesting the effectiveness of the domain prediction branch in adaptively weighing source domains contributions. DGER (w/ Max-Voting) generally outperforms DGER (w/ Averaging), especially in heterogeneous domains, aligning with the expectation that selecting the most relevant domain is more effective when domain divergence is significant. Notably, in certain heterogeneous domains, such as Walmart-Amazon and Amazon-Google, DGER (w/ Averaging) performs worse than the best baseline, highlighting the risk of directly averaging source domains when domain heterogeneity is high, as it may introduce more noise and irrelevant information from less similar source domains. This ablation study emphasizes the importance of selecting the appropriate inference strategy based on domain characteristics and provides guidance for designing effective DG approaches in ER.

### 6.2.3. Fine-tuning DGER with DUAL

In this section, we experimentally validate the effectiveness of the DUAL framework in fine-tuning the DGER model. Our experimental design reflects the three core principles outlined in Section 4: uncertainty-based sampling, domain-aware uncertainty sampling, and pseudo-label enhanced iterative learning. The experiments commenced

with a predetermined annotation budget of 200, signifying the allocation of 200 samples for annotation. The sampling granularity, indicative of the number of samples selected in each active learning iteration, was fixed at 20. Three sets of experiments were conducted on each of the three datasets: Abt-Buy, Walmart-Amazon, and Amazon-Google.

#### (1) Uncertainty Sampling vs. Random Sampling

The initial experiments focused on comparing active learning with uncertainty-based sampling against random sampling. The objective was to ascertain if fine-tuning the DGER model using uncertainty-based samples resulted in superior generalization performance on the target domain.

Fig. 9 illustrates that by actively querying samples for annotation, the uncertainty-based approach outperforms random sampling, achieving higher model performance with the same annotation budget. Moreover, uncertainty-based sampling generally exhibits more stable performance improvements (as shaded areas) compared to random sampling. This showcases the core essence of active learning in strategically guiding the DGER model to better adapt to the target domain. However, this method neglects a crucial aspect: the divergence between the sample and the source domain, highlighting the need for deeper exploration.

#### (2) Domain-Aware Uncertainty Sampling vs. Uncertainty Sampling

Building on our initial findings, this set of experiments evaluated the effect of integrating a domain-aware factor during the selection of highly informative samples. Specifically, we explored whether combining sample uncertainty with source divergence would enhance the DGER model's fine-tuning process over uncertainty sampling alone.

Fig. 10 demonstrates the advantages of incorporating domain awareness into uncertainty sampling, a standout feature of the DUAL

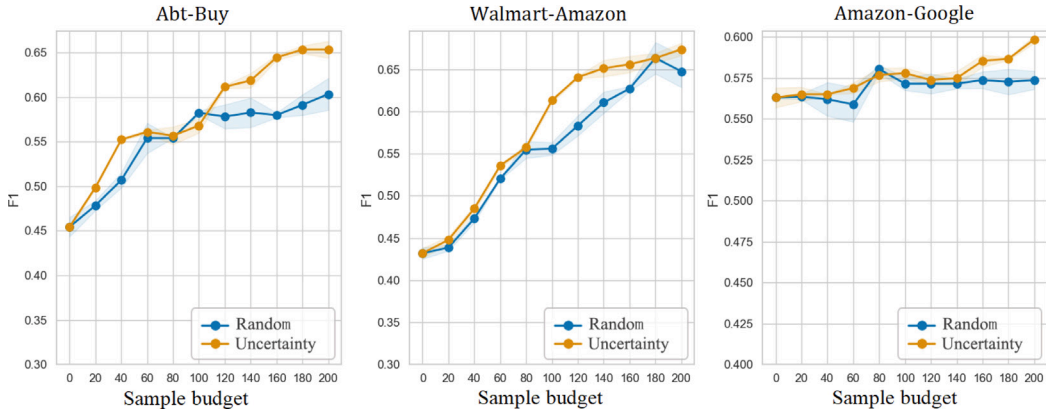


Fig. 9. Performance comparison between uncertainty-based and random sampling methods on the Abt-Buy, Walmart-Amazon, and Amazon-Google datasets. With an increase in the number of annotated samples (x-axis), uncertainty-based sampling typically yields higher F1 scores (y-axis) than random sampling.

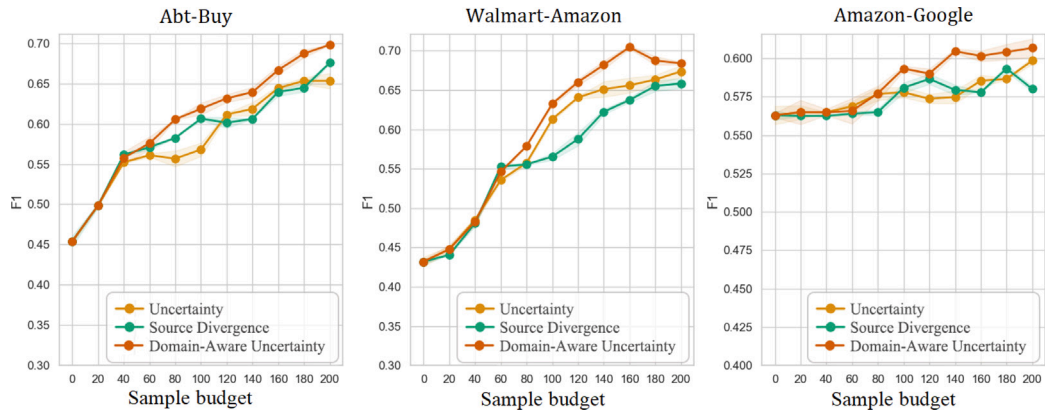


Fig. 10. Performance comparison between domain-aware uncertainty sampling, uncertainty sampling, and source divergence sampling on the three datasets. The integration of sample uncertainty and source divergence further improves the fine-tuning process, resulting in higher F1 scores for the same annotation budget.

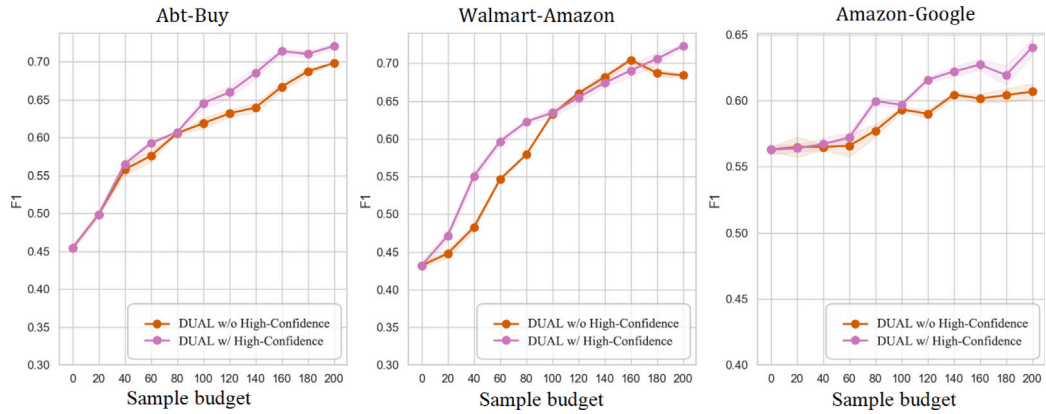


Fig. 11. Effect of including high-confidence samples on DUAL performance across the three datasets. Integrating high-confidence samples into the DUAL framework tends to enhance the model's performance on the target domain, as reflected by higher F1 scores with the same number of annotated samples.

framework. By considering samples' source divergence, the fine-tuning process of the DGER model can be significantly enhanced. This underscores the significance of domain-aware uncertainty sampling in optimizing model performance within a limited annotation budget.

### (3) DUAL With vs. Without High-Confidence Samples

In the final set of experiments, we examined the impact of integrating high-confidence samples with pseudo-labels into the DUAL

framework. The premise was that leveraging unlabeled data through high-confidence samples would boost the DGER model's performance.

Fig. 11 highlights the impact of incorporating high-confidence samples with pseudo-labels into the fine-tuning process. As shown, introducing high-confidence samples significantly boosts the DGER model's performance in the target domain, underscoring the benefits of leveraging unlabeled data with pseudo-labels in the DUAL framework.

Throughout the experiments, the efficacy of the DUAL strategy in fine-tuning the DGER model becomes increasingly evident. Integrating active learning with domain-aware uncertainty sampling and leveraging high-confidence samples in the learning process leads to enhanced performance in the target domain, while simultaneously reducing the need for annotations.

## 7. Conclusion

This work presents a novel two-phase framework to address the low-resource challenge in ER. The first phase, DGER, aims to train an ER model on multiple source domains that can generalize well to unseen target domains. It combines domain adversarial learning and simulated target learning to improve the model's out-of-distribution generalization ability. To further apply the DGER model to a specific challenging target domain, the second phase, DUAL, is proposed as an innovative active learning strategy for fine-tuning the DGER model. DUAL efficiently selects the most informative samples, considering both uncertainty and divergence from the source domain, along with high-confidence samples for pseudo-labeling. This approach effectively fine-tunes the DGER model while optimizing the annotation budget. Our experimental results demonstrate the superiority of our framework over traditional ER methods in generalizing to unseen domains. Furthermore, our fine-tuning strategy outperforms traditional uncertainty-based active learning techniques within a constrained annotation budget. This research contributes to the development of more efficient ER systems, with potential applications across various domains, especially in low-resource settings. In future work, we aim to enhance DGER framework's efficiency by exploring lightweight strategies for reduced training time. Additionally, we will address dataset diversity and fairness issues, particularly focusing on mitigating biases due to unequal dataset sizes, to ensure generalization across diverse domains.

## CRedit authorship contribution statement

**Zhihong Xu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ning Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

All authors declare that No conflict of interest exists.

## Data availability

Data will be made available on request.

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